

23 C In electric field

$$\frac{1}{2}mv^2 = qV$$

$$v = \sqrt{\frac{2qV}{m}}$$

In magnetic field,
electromagnetic force provides centripetal force

$$Bqv = \frac{mv^2}{r}$$

$$Bq = \frac{mv}{r} = \frac{m}{r} \sqrt{\frac{2qV}{m}}$$

$$\frac{m}{q} = \frac{B^2 r^2}{2V}$$

$$\therefore \frac{m}{q} \propto r^2$$

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$$\frac{\alpha_x}{\alpha_y} = \left(\frac{L}{2L}\right)^2 = \frac{1}{4}$$

24 D Final resultant magnetic flux density =